Exam AWS Certified Developer - Associate DVA-C02 All Questions

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EXAM AWS CERTIFIED DEVELOPER - ASSOCIATE DVA-C02 TOPIC 1 QUESTIO...

Exam question from Amazon's AWS Certified Developer - Associate DVA-C02
Question #: 74
Topic #: 1
[All AWS Certified Developer - Associate DVA-C02 Questions]

A developer created an AWS Lambda function that accesses resources in a VPC. The Lambda function polls an Amazon Simple Queue Service (Amazon SQS) queue for new messages through a VPC endpoint. Then the function calculates a rolling average of the numeric values that are contained in the messages. After initial tests of the Lambda function, the developer found that the value of the rolling average that the function returned was not accurate.

How can the developer ensure that the function calculates an accurate rolling average?

A. Set the function's reserved concurrency to 1. Calculate the rolling average in the function. Store the calculated rolling average in Amazon ElastiCache.

B. Modify the function to store the values in Amazon ElastiCache. When the function initializes, use the previous values from the cache to calculate the rolling average.

C. Set the function's provisioned concurrency to 1. Calculate the rolling average in the function. Store the calculated rolling average in Amazon ElastiCache.

D. Modify the function to store the values in the function's layers. When the function initializes, use the previously stored values to calculate the rolling average.

Show Suggested Answer

by Fyssy at *April 19, 2023, 1:43 p.m.*

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Comments

eboehm Highly Voted 2 years, 9 months ago

Selected Answer: A

You need to set the reserved concurrency to 1 otherwise multiple functions could run at the same time causing the math to be off. Also there was a similar question in another practice exam set that stated the same thing

upvoted 25 times

jipark 2 years, 8 months ago

reserve concurrency 1 means poll in order.
this looks answer.

upvoted 2 times

MrTee Highly Voted 2 years, 11 months ago

Selected Answer: B

By using ElastiCache, the Lambda function can store the values of the previous messages it has received, which can be used to calculate an accurate rolling average.

upvoted 19 times

sumanshu Most Recent 1 year, 3 months ago

Selected Answer: A

B) Eliminated - If multiple instances of the Lambda function run concurrently, they could retrieve or modify the same data in ElastiCache, leading to inaccuracies in the rolling average.

C) Eliminated - Provisioned concurrency does not prevent concurrency issues

D) Eliminated - Lambda layers are used to share common code or dependencies across Lambda functions but are not designed for storing or persisting data.

upvoted 2 times

ShakthiGCP 1 year, 3 months ago

Selected Answer: B

Reserved concurrency: Setting reserved concurrency to 1 ensures only one instance of the function executes at a time. While this might prevent race conditions, it doesn't address the core issue of calculating the rolling average across multiple Lambda invocations.
u

upvoted 1 times

trieudo 1 year, 3 months ago

Selected Answer: A

==> Discard B: There are not limit corrency in elastic cache, multiple lambda can access make race condition ==> wrong output

upvoted 1 times

AmitRanchi 1 year, 5 months ago

Selected Answer: A

Reserved Concurrency of 1 ensures that only one instance of the Lambda function runs at a time, which is crucial for calculating an accurate rolling average. If multiple instances of the Lambda function run simultaneously (without concurrency control), each instance might have incomplete or inconsistent data, leading to incorrect results for the rolling average.

Storing the rolling average in Amazon ElastiCache provides a persistent, in-memory storage that the function can access quickly during each invocation. ElastiCache can store the current state of the rolling average and any relevant historical data, ensuring that each invocation of the Lambda function has access to the most recent values for calculation

upvoted 1 times

Saurabh04 1 year, 8 months ago

Correct answer should be D. Storing values in the function's layers and using them during initialization is straightforward. This Approach avoids external services like Elasticache.

upvoted 1 times

rue_ 1 year, 5 months ago

layers are not meant to store dynamic data. it's for sharing libraries and static code across lambda functions

upvoted 1 times

lozou 1 year, 8 months ago

Selected Answer: B

elasticache for keeping the values from previous invocation

upvoted 1 times

guidosolano 1 year, 9 months ago

Selected Answer: B

A y C limitan mucho la capacidad de Lamda. Voy con B

upvoted 1 times

65703c1 1 year, 10 months ago

Selected Answer: A

A is the correct answer.

upvoted 2 times

SathyaJS 2 years ago

Selected Answer: B
As per Gemini below is why A is incorrect
A. Reserved concurrency: Setting reserved concurrency to 1 ensures only one instance of the function executes at a time. While this might prevent race conditions, it doesn't address the core issue of calculating the rolling average across multiple Lambda invocations.

upvoted 1 times

yingying920928 2 years ago

Selected Answer: B

While limiting concurrency (A or C) can help manage the function's execution rate and scale and prevent throttling, it is not directly related to ensuring the accuracy of calculating rolling average. Instead, focusing on proper state management and data consistency mechanisms (using ElastiCache) is key to achieving accurate results in this scenario.

upvoted 2 times

maurice2005 2 years ago

that's the correct way in real world but nothing about state management or data consistency is mentioned in B. But A has it although it's not the good real world to do so.

upvoted 1 times

SerialiDr 2 years, 1 month ago

Selected Answer: A

Option A ("Set the function's reserved concurrency to 1. Calculate the rolling average in the function. Store the calculated rolling average in Amazon ElastiCache.") is the most suitable solution. It ensures that only one instance of the Lambda function processes messages at any given time, maintaining the sequence of message processing which is crucial for an accurate rolling average calculation. Additionally, using Amazon ElastiCache to store and retrieve the rolling average across invocations addresses the statelessness of AWS Lambda, enabling stateful processing.

upvoted 4 times

d323bvmiqj 2 years, 1 month ago

Selected Answer: A

What if one of the instances freezes and holds one of the vzlues for some time, not updating cache, while the others continue calculatinv the avg giving wrong output ?

upvoted 3 times

SerialiDr 2 years, 2 months ago

Selected Answer: B

By storing individual message values in ElastiCache (a fast, in-memory data store), the Lambda function can retrieve these values upon initialization to calculate an accurate rolling average. This approach effectively maintains state across Lambda invocations.

upvoted 2 times

Chimzi 2 years, 2 months ago

Selected Answer: B

Using ElastiCache allows you to maintain a shared state across all instances of your Lambda function

upvoted 1 times

ShinobiGrappler 2 years, 3 months ago

Selected Answer: A

This approach controls concurrency by ensuring only one instance runs at a time. Provisioned concurrency also has the added benefit of reducing cold start latency. Storing the rolling average in ElastiCache is a good practice for maintaining state. However, like option A, it may limit the function's throughput.

upvoted 4 times